DO NOW — RECALL

Lesson 060 · Refining an Abstraction and Testing Assum

Name: _____ Class: _____ Date: _____

1. What does abstraction remove?

2. What is a model assumption?

3. What is a boundary test?

4. Two rising-edge D-type flip-flops A and B share a clock and start with QA=0 and QB=0. DA=1 throughout, and DB is connected to QA. Explain why QB is still 0 immediately after the first rising edge. Assume ideal simultaneous sampling.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Irrelevant details.

2. Model answer

Simplifying rule treated as true for a task.

3. Model answer

An input at/around an important threshold.

4. Model answer

Both sample their inputs at the same edge. B samples the old QA=0, while A samples 1. A's new output reaches B after that sampling event.

By the end, you can…

• Compare the simplification against new requirements. If the model omits a relevant factor it can produce plausible but wrong outputs.

• Refine the abstraction by adding only the newly necessary variable, rather than reconstructing every real-world detail.

• Test normal, boundary and exceptional cases: 5 km versus 6 km, return and single, and invalid negative distance.

Vocabulary — know these terms

refinement — Improving model for new requirements.

boundary — Important threshold or edge.

validity — Whether model works for its intended use.

assumption — Rule taken as given.

NEW LEARNING

Read, annotate and explain

Understand

Compare the simplification against new requirements. If the model omits a relevant factor it can produce plausible but wrong outputs.

Model

Refine the abstraction by adding only the newly necessary variable, rather than reconstructing every real-world detail.

Evaluate

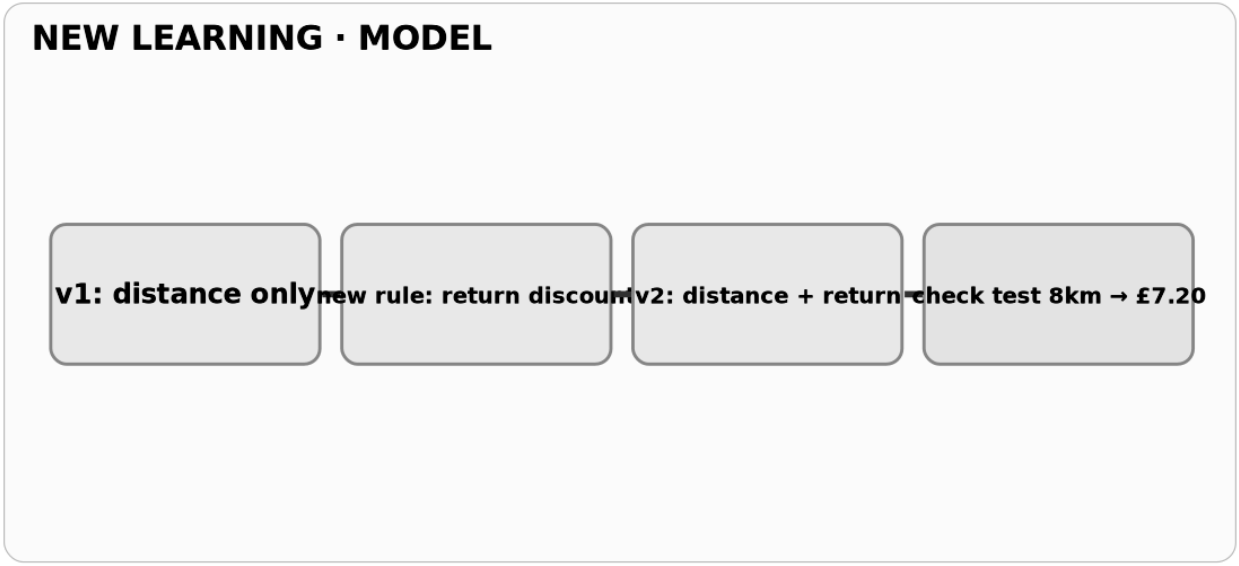
Test normal, boundary and exceptional cases: 5 km versus 6 km, return and single, and invalid negative distance.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · MODEL: v1: distance only; new rule: return discount; v2: distance + return; check test 8km → £7.20

I DO — WORKED EXAMPLE

Follow the worked method

A simplified cinema ticket model uses only age but now weekend prices differ. What change is needed?

Explain

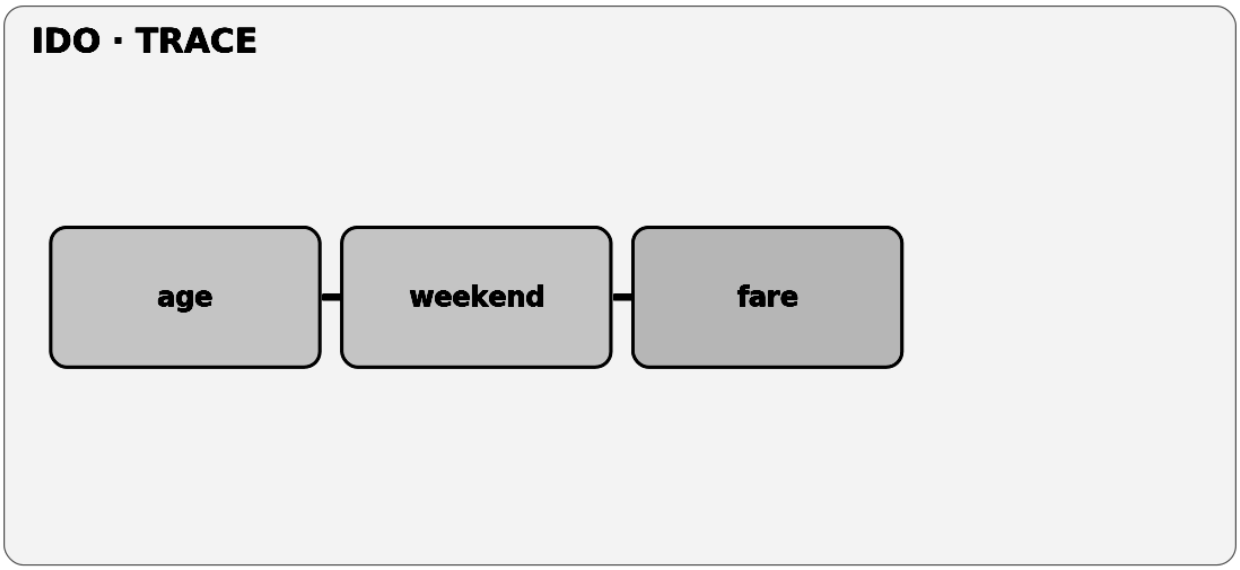
Weekend becomes a relevant distinction because it can change the output.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A simplified cinema ticket model uses only age but now weekend prices differ. What change is needed?

Weekend becomes a relevant distinction because it can change the output.



IDO · TRACE: age; weekend; fare

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A parking price model charges £3/hour but introduces £12 maximum. Improve algorithm.
Source code / input:
fee=MIN(3×hours,12), with validation that hours≥0.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Ticket v1 charges £2 at ≤5km, £4 otherwise. Give charge at 5km and 6km and why these are useful tests. [6]

Question 2

Return journeys cost 1.8 times single. For 8km journey calculate return fare and show variables. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

A school model predicts homework concerns from just missed work counts; discuss which relevant conditions to add and how to test impact. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A model that worked yesterday never needs changing when requirements change.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

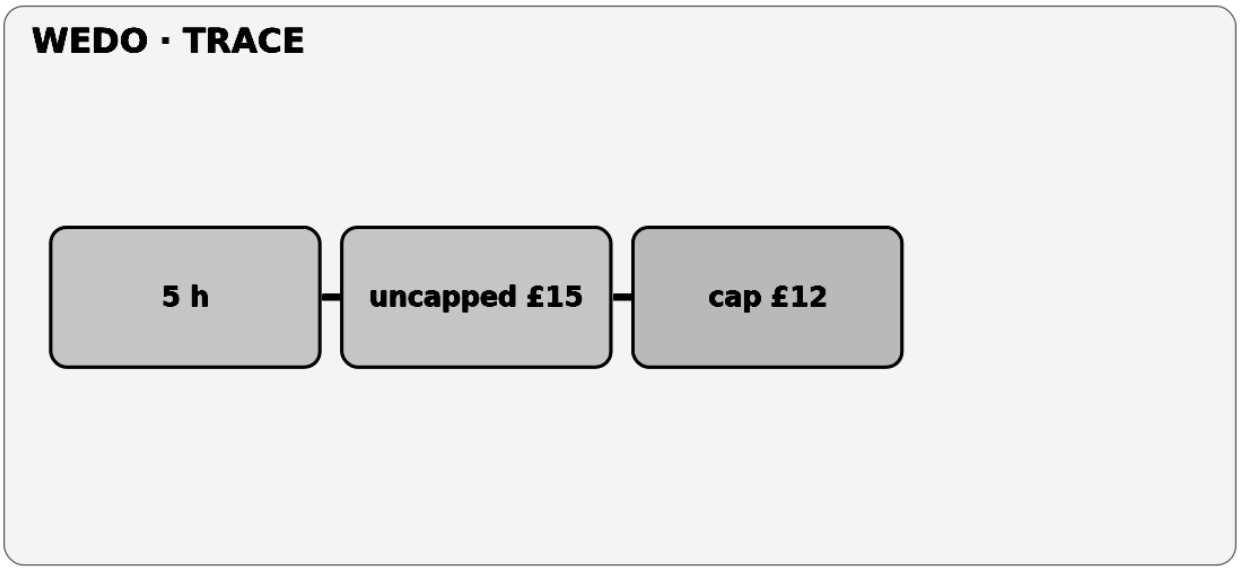
Explain
Cap is new relevant rule. Example 5 hours costs £12, not £15.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A parking price model charges £3/hour but introduces £12 maximum. Improve algorithm.

Cap is new relevant rule. Example 5 hours costs £12, not £15.



WEDO · TRACE: 5 h; uncapped £15; cap £12